

What can the agent do now that it could not do six sessions ago?

Six sessions ago, the agent could read a financial statement and comment on the reported numbers, but it did not have a consistent framework for turning those numbers into analytical statements. It can now start from the annual report, extract the reported figures, trace material lines into the notes, and independently apply our predefined classification rules. It distinguishes core operating, non-core operating, financial and unusual items, produces the three operating-cost reformulations, and links income-statement classifications to the managerial balance sheet. Most importantly, it now has explicit rules for what to do when disclosures are insufficient rather than filling gaps with assumptions.

Which methodological choice was the hardest, and why did you decide as you did?

The hardest methodological choice was defining what we considered unusual. Rheinmetall itself reports special items and adjusts its operating result for both special items and PPA effects, but simply adopting management's adjustments would have removed the analytical judgement the exercise is designed to develop. We therefore defined unusual items independently using the criteria from the course: nature, frequency, matching, discontinued operations and complex valuation effects. We also distinguished between an account that is unusual in its entirety and an unusual change within an otherwise recurring account. This became particularly relevant for provision reversals, disposal gains and losses, impairment-related items and transaction effects, which cannot automatically be treated as non-recurring simply because they appear unusual by name.

We chose not to classify PPA amortisation automatically as an unusual item. Instead, we retain it in the primary EBIT-based reformulation and analyse EBITA separately. This keeps the reported economics visible while allowing us to assess the acquisition-related amortisation separately.

Another important development concerned the first operating-cost reformulation, the income statement by function. Rheinmetall reports its operating expenses primarily by nature rather than by function, meaning that costs such as materials, personnel, depreciation and other operating expenses are disclosed, but we do not have a sufficiently reliable allocation between cost of sales, selling expenses, G&A and R&D. We therefore decided not to manufacture a Cost of Sales or Gross Profit figure using arbitrary allocations. This follows the course material, which warns that allocating nature-based costs to functions can require considerable judgement. At the same time, we realised that simply writing “not determinable for Rheinmetall” into the prompt would make the agent company-specific. We therefore added a general decision rule to Module A. The agent must first identify whether the company reports expenses by function or by nature. If a reliable functional statement or sufficient note disclosures exist, it should construct the functional reformulation. For a manufacturing company, it should test whether direct materials, direct labour, factory overhead and inventory movements are available to derive Cost of Sales. For a retail company, it should instead use merchandise purchases adjusted for beginning and ending inventory. If these inputs cannot be established without material assumptions, the agent must report Cost of Sales and Gross Profit as not reliably determinable rather than inventing an allocation. These manufacturing and retail approaches follow the methodology given in the course.

Where did the agent mislead you, and how did you catch it?

One of the clearest cases where the agent misled us was during the extraction of reported figures. It reconstructed certain headline accounts by summing the rounded figures disclosed in the notes. This looked reasonable and the individual components were correct, but the resulting totals did not always equal Rheinmetall's reported figures. We caught the issue by reconciling every extracted headline against the face of the annual report rather than relying only on the note breakdown. We subsequently changed both the workbook and the prompt: the reported headline must now be taken exactly from the primary statement, while note components are reconciled separately through an explicit rounding/residual line. This was a useful reminder that an internally logical calculation can still be factually wrong.

Another important error concerned our calculation of net financial obligations (NFO). The agent initially deducted all non-operating assets from financial debt, including equity-method investments and assets held for sale. When we reviewed the course slides, we realised that NFO should deduct financial assets only. We therefore separated non-operating assets into non-core operating assets, held-for-sale assets and financial assets, and recalculated NFO. This also changed the Modigliani–Miller and Penman ROE decompositions. We added the distinction explicitly to the prompt and introduced reconciliation checks so the same mistake should not occur again.